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Enhancing Antibiotic Effectiveness:
An Introduction to Sulbactam and Related β-Lactamase Inhibitors

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PHS1101

INTRODUCTION TO THE DEVELOPMENT AND STRUCTURE OF SULBACTAM

History of β-lactamase Inhibitors Development

1980s – Sulbactam was introduced to inhibit class A β-lactamases and is commonly combined with Ampicillin to treat resistant infections [1]

- **1928** – Sir Fleming discovered the first antibiotic penincillin
- **1940** – First case of penicillin resistance via penicilinase reported
- **1962** – The last class of broad-spectrum antibiotics is discovered
- **1970** – Clavulanic acid was developed as the first β-lactamase inhibitor, restoring antibiotic efficiency [1]
- **1980** – Vancomycin-Resistant Enterococcus (VRE) Discovered
- **1984** – Amoxicillin/ Clavulanate is approved for clinical use
- **1987** – Sulbactam/ Amipicillin is approved for clinical use
- **Now** – Ongoing research on novel β-lactamase Inhibitors in the face of antimicrobial resistance as a global threat

Key Functional Group AND Predicted intermolecular Interactions with the Target

Note: HBA: Hydrogen bond acceptor

Amide Linkage (-N-C=O)

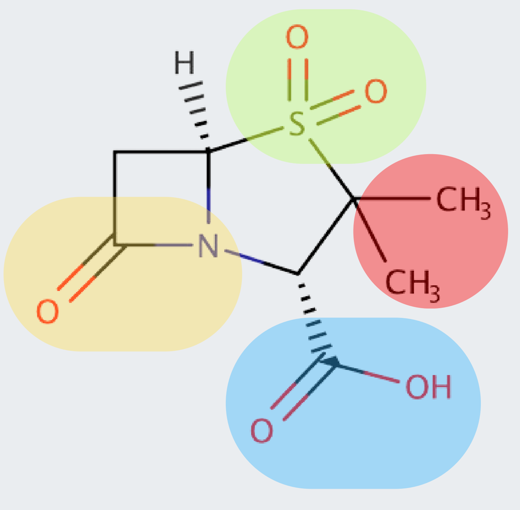
Hydrogen Bonding between **N/O (HBA)** and **polar amino acids** (e.g. Ser).

Carboxylic Group (-COOH)

Ion-Ion interactions between **ionized COO^-** with **basic amino acids** (e.g. Arg/Lys).

Hydrogen Bonding between **O (HBA)** and **polar amino acids** (e.g. Ser/ Thr).

Remarks: The ionization depends on the local pH of the physiological environment, affecting its interactions with the enzyme.



Sulbactam

Sulfonyl Group ($-SO_2$)

Ion-Dipole Interactions between **polar SO_2 group** and **charged amino acids**(e.g. Arg).

Hydrogen Bonding between **O (HBA)** and **polar amino acids** (e.g. Ser).

Remarks: These interactions may stabilize the binding of Sulbactam at the active site.

Methyl Groups (CH_3)

In addition, the **methyl groups** may form **hydrophobic interactions** with **non-polar amino acid residues** (e.g. Ala), increasing the stability of Sulbactam binding.

Key Interactions with the Environment

Sulbactam's polar groups($-SO_2$, -COOH) forms hydrogen bonds with water molecules **which improves its solubility and enables effective distribution to the aqueous physiological environment** to reach and bind to β-lactamases.

β-lactam Ring and Covalent Bond Formation

The β-lactam ring with **δ^+** carbon is susceptible to nucleophilic attack by the active-site **serine residue** (illustrated in Fig.5). This leads to the **opening of the β-lactam ring** and the formation of an **acyl-enzyme intermediate**, leading to the inactivation of β-lactamase.

DRUG-LIKENESS EVALUATION OF SPECIFIC DRUGS

Drug	Lipinski's Rule of Five				
	Molecular weight (<500)	LogP(<5)	HBA Count(<10)	HBD Count(<5)	Whether Aligned
Sulbactam	233.24	-0.89	5	1	YES
Clavulanate	199.16	-1.2	5	2	YES
Tazobactam	300.29	-1.4	7	1	YES
Enmetazobactam	314.32	-4.6	6	0	YES

Drug	Veber's Rules			Lovering's Rule	
	Polar Surface Area (<140 Å^2)	Rotatable bond count(≤ 10)	Whether Aligned	Fraction of sp^3 (>0.40)	Whether Aligned
Sulbactam	91.75	1	YES	0.75	YES
Clavulanate	87.07	2	YES	0.50	YES
Tazobactam	122.46	3	YES	0.60	YES
Enmetazobactam	116.28	3	YES	0.64	YES

All data included in this table are taken from DrugBank, except for Fsp3 values (manually calculated)

STRUCTURAL-PHARMACOKINETIC RELATIONSHIPS (I)

Drug	Route of Absorption *IV: Intravenous* *IM: Intramuscular*	Oral bioavailability
Sulbactam Tazobactam Enmetazobactam	Parenteral – IV or IM injections 	**Poor** The **strained β-lactam ring structure** in drugs make it **more prone to opening** and **susceptible to degradation** by proteases and acidic environment in the gastrointestinal tract. All drugs are **hydrophilic**, with **low lipophilicity**. They have **limited ability** to pass through hydrophobic cell membranes of intestinal cells to enter circulation.
Clavulanate	Enteral – taken orally as tablet/ liquid suspension 	**Good** Clavulanate is **less hydrophilic** than sulbactam (lack of sulfone group). Clavulanate is **stable** in **acidic pH** of stomach. The presence of **electron-withdrawing groups** in the side chains **stabilise** the β-lactam ring, **reducing** the likelihood of **nucleophilic attack** by water molecules. Presence of **oxazolidine structure** contributes to clavulanate's **resilience** to **hydrolytic degradation** in acidic environments.

DRUG-TARGET INTERACTIONS AND MECHANISM OF ACTION

Sulbactam acts as a competitive, irreversible inhibitor of class A β-lactamases [2]. Known as serine β-lactamases, they rely on **serine at the active site for hydrolysis of β-lactam antibiotics** (Fig. 5). Fig. 6 shows the binding of sulbactam to the active site of SHV-1 β-lactamase. Sulbactam carboxyl group interacts with important residues like **R244, S130, and T235 via hydrogen bonds** with its **carbonyl oxygen positioned in the oxyanion hole** [4]. These interactions show how Sulbactam blocks the enzyme's active site.

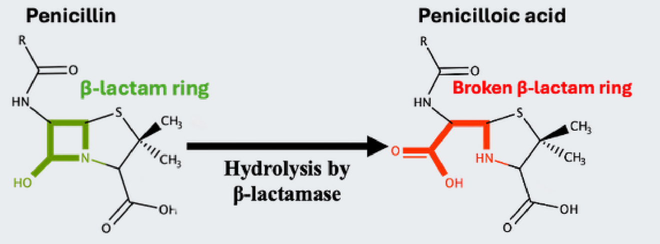
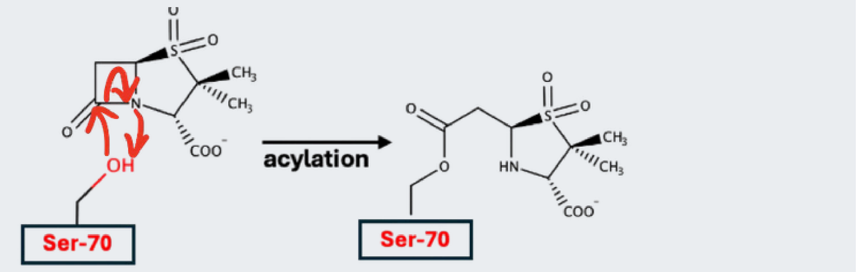
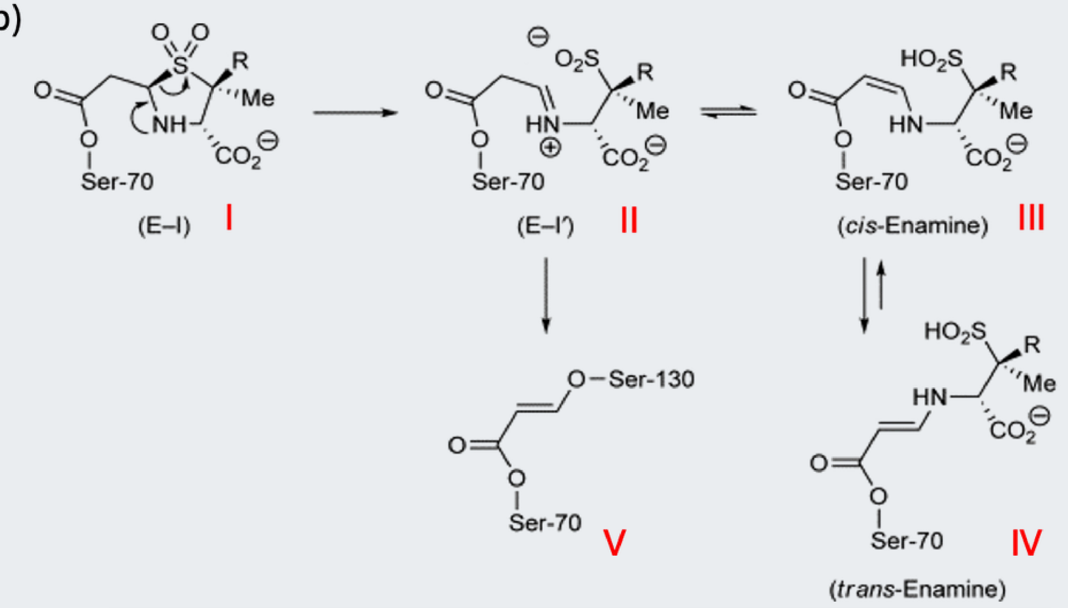


Fig. 5 Mechanism of penicillin resistance through hydrolysis of the β-lactam ring by β-lactamase.

(a)



(b)



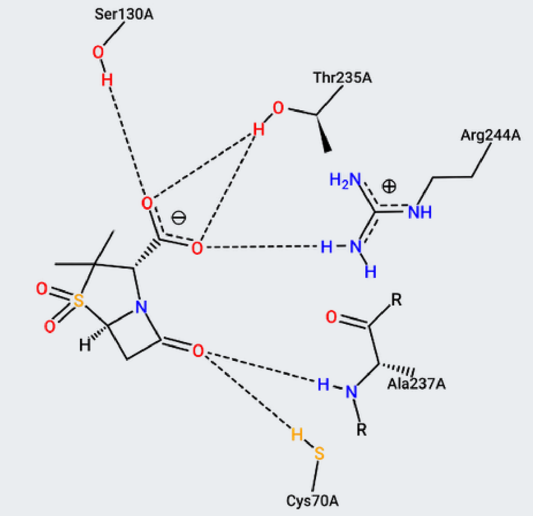


Fig. 6 Interaction map between sulbactam and S70C variant of SHV-1 β-lactamase. To capture the pre-acylation complex, serine 70 of the enzyme is mutated to cysteine to slow down the acylation reaction [4].

Fig. 7(a) Formation of an acyl enzyme intermediate via nucleophilic attack by serine on the β-lactam ring of Sulbactam,. (b) Inhibition pathways of class A β-lactamases by sulbactam [5]. Structure V shows the final state of irreversible inhibition via covalent modification at position Ser-130.

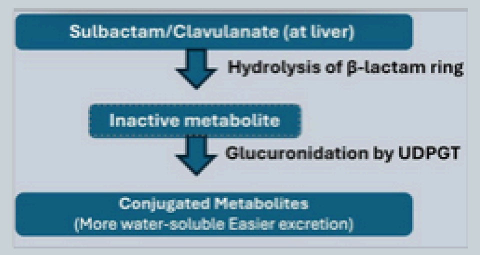
Fig. 6 is generated using ProteinPlus PoseView (Protein PDB-code: 4FH2). Fig. 6(a) chemical structures are computed using Marvin 24.3.0 .

The mechanism of action relies on the **formation of acyl-enzyme intermediate** (Fig. 7a). Subsequently, the second ring opens in forming **an imine intermediate (structure II)** (Fig. 7b). The imine then isomerises to cis-enamine (structure III), which can isomerize to trans-enamine (structure IV). **This interconversion via isomerisation results in transient inhibition.** After several hours, upon **covalent modification by Ser-130, irreversible inhibition is achieved (structure V)**, where there is a permanent disruption to the enzymes' active site, making it unavailable for hydrolysing β-lactam ring of the antibiotics [3]. **This restores the antibiotics' efficacy** in binding to Penicillin Binding Proteins of the bacteria.

Sulbactam's Binding to β-Lactamases vs Antibiotic's Binding to β-Lactamase

Similarities	Differences
Involves the intermolecular interactions (e.g. hydrogen bonding) between polar functional groups and nearby polar amino acids.	Enzyme's catalytic efficiencies for hydrolyzing Sulbactam is much lower due to processes seen in Fig. 7(b) and the **active site is eventually permanently inactivated.**
Involves the hydrolysis of **β-lactam ring of Sulbactam/Penicillin** by **Serine residue** at the active site.	In contrast, enzymes are highly efficient at hydrolyzing Penicillin and the hydrolyzed product quickly leaves, leading to **regeneration of the active site.**

STRUCTURAL-PHARMACOKINETIC RELATIONSHIPS (II)

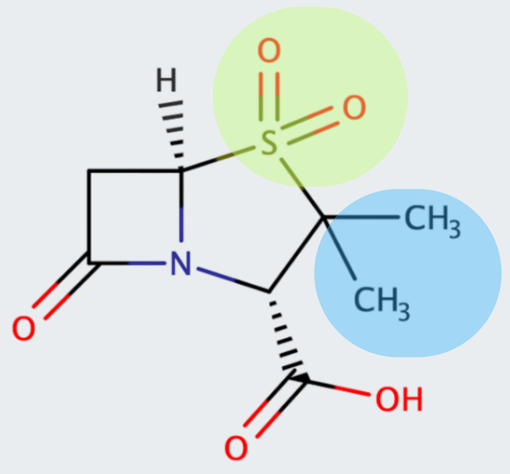
Drug Generic Name	Metabolic Clearance	Renal Clearance		Half-life
Sulbactam		~12L/h	**Remark:** Having low LogP, Sulbactam is **relatively hydrophilic**; • Hence, it is well absorbed into the blood and primarily eliminated by mainly **glomerular filtration**. When renal clearance exceeds the glomerular filtration rate, it is subject to **tubular secretion**. • Minor elimination via biliary secretion.	~ 1h
Clavulanate		12.2L/h	**Remark:** Clavulanate's **LogP is higher** than sulbactam. Being less hydrophilic and less distribution in blood, its renal clearance is also lower.	~ 0.8h
Tazobactam		~3-5L/h	**Remark: Eliminated by renal elimination** with 80% of drug remaining unchanged. It is a substrate of the transporters **OAT1, OAT3**, inhibitors of these transporters should be avoided.	0.7 h – 1.2h
Enmetazobactam	****Minimally** metabolised, prolonging its activity. **Resists hydrolysis** as the additional methyl group **hinders access** to the **active site** of enzymes, prevents β-lactam ring from being opened.	7.6L/h	**Remark:** Enmetazobactam is eliminated by glomerular filtration and tubular secretion. About 90% of Enmetazobactam is excreted unchanged in the urine	2.6h

Renal Clearance data included in this table are taken from DrugBank.

STRUCTURAL COMPARISON WITH SAME CLASS DRUGS IN RELATION TO PROPERTIES

Fig.1-4 and structure of Avibactam are computed using MarvinSketch 24.3.0.

Fig 1. Sulbactam

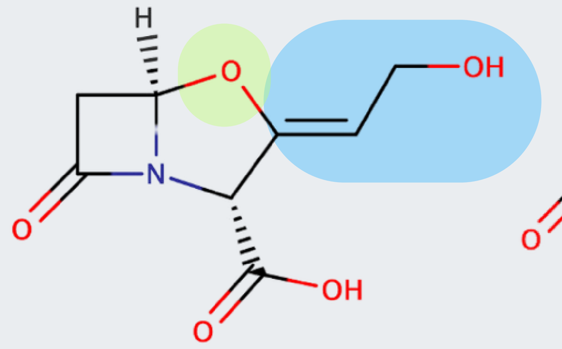


The **polar sulfone group** forms **dipole-dipole interactions/hydrogen bonds** with **polar amino acids** at the pocket.

The non-polar CH_3 groups in Sulbactam:
- Increases lipophilicity
- Increases membrane permeability
- Decreases water solubility

Solubility of sulbactam: 48.5 mg/mL

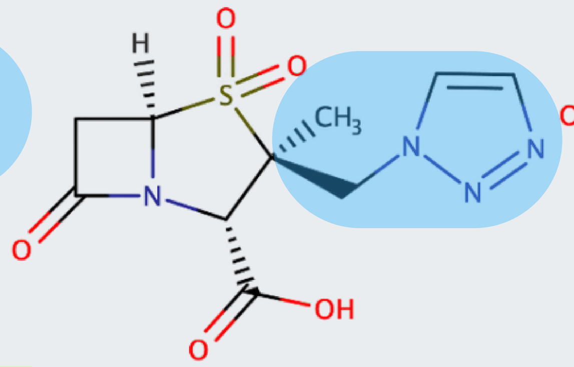
Fig 2. Clavulanate



Despite being slightly **more polar,** the **sulfone group** is a much bulkier group compared with **oxygen atom** in **Clavulanate,** this causes clavulanate to have a smaller molecular size and therefore being more soluble.

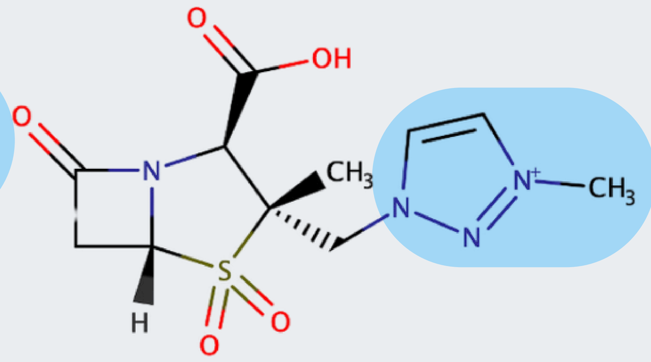
The polar hydroxyl group in **Clavulanate** may account for the **lower lipophilicity** than **Sulbactam** which enables it to **form H bond** with **water** and **polar AA.**
The greater polarity coupled with a **smaller molecular size** contributes to its **higher solubility (337.0mg/mL)** compared with **sulbactam.**

Fig 3. Tazobactam



The triazole ring in **Tazobactam** is nitrogen rich and polar which contributes to an even **lower lipophilicity** compared with **Sulbactam.**
Its **high melting point** and a **large molecular size** may counteract increased polarity to give a **similar solubility (50.0mg/mL)** as sulbactam.

Fig 4. Enmetazobactam



The **additional methyl group** in **Enmetazobactam** forms a **positively-charged quaternary amine,** increasing **water solubility** due to presence of **triazolium ion.**

This may **lower** the **drug permeability** due to the resulting **low lipophilicity.**

Common Structural Features

β-Lactam Ring	Enables the drugs to act as a **suicide inhibitor** through mimicking β-lactam antibiotics, covalently binding to the enzyme's active site **upon hydrolysis of the ring.**
Carboxyl Group	The ionized -COO⁻ leads to improved binding by forming **ion-ion interactions/ion-dipole interactions** with charged/polar amino acids. • It may help to orientate the molecule for better binding by interacting with specific set of amino acids at the pocket (seen in Fig. 6). • **The carboxyl group, especially in charged form, may ensure the drugs have good water solubility.**

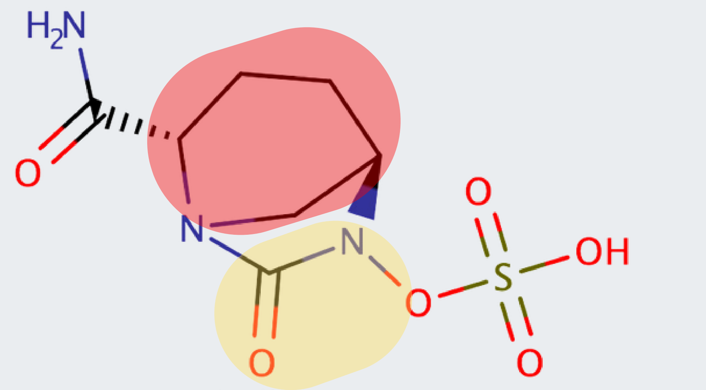
	Ionisable Groups	Discussions on Flexibility		
Sulbactam	**Carboxylic acid**	Fused rings, β-lactam ring, thiazolidine, reduce flexibility.	Sulfone group does not increase flexibility. Overall sulbactam is a relatively rigid molecule.	
Clavulanic Acid			Lack of bulky side chains (sulfone) makes clavulanate slightly more flexible, but still rigid.	
Tazobactam			Sulfone group does not increase flexibility.	The triazole ring in tazobactam provides some flexibility compared to a fused ring core, but overall, it remains rigid.
Enmetazobactam	**Carboxylic acid and N-methyl triazolium ion** *			The methyl group attached to the triazole increases flexibility as it is a simple alkyl chain that can rotate freely, making enmetazobactam more flexible than tazobactam.

* The **additional N-methyl triazolium ion increases water solubility** in **acidic conditions** (e.g. stomach). Enmetazobactam's **basic groups** may allow for **stronger ionic interactions** with **negatively charged amino acid** in β-lactamase enzymes, potentially improving its inhibitory action in certain environments.

Extension : Second Generation β-lactamase inhibitor- Avibactam as an Example

Despite not within the main scope of discussion, it is interesting to note that there are **β-lactamase inhibitors that do not contain β-lactam rings**(e.g. Avibactam, as seen in Figure below).

The diazabicyclooctane (DBO) scaffold allows avibactam to interact with active sites of a greater range of β-lactamases (class A, C and some class D enzymes) [6]. The DBO scaffold may interact better with non-polar amino acids via hydrophobic interactions which allows it to accommodate active sites of greater diversity.



Avibactam

The -N-C=O is present in a five-membered ring rather than a four membered ring. With less ring strain in a five-member ring, after acylation, it is now possible for avibactam to deacylate and recyclize [6]. This leads to its unique mechanism of reversible inhibition in contrast to suicide inhibition.